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A pressure-based numerical scheme for compressible–incompressible two-phase flows

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Abstract

This article presents a numerical scheme for two-phase flows consisting of compressible gas and incompressible liquid. Assuming that the gas density is determined by pressure only and that the liquid density is constant, we develop a new set of governing equations for compressible–incompressible two-phase flows, which can be seen as a simplification of the previous unified governing equations. The volume fraction of gas, velocity, and pressure are the primary variables. Given the liquid's usual leading role in the flow motion, a pressure-based method is employed to solve the equations. A fifth-order accurate weighted essentially nonoscillatory (WENO) scheme is applied to discretize the advection terms, and the tangent of hyperbola for interface capturing (THINC) scheme is utilized to capture the interface. The numerical scheme proposed is validated against the generalized Bagnold model and the free drop of a water patch in a closed tank. The results show that the scheme can tackle low Mach number compressible–incompressible two-phase flows. For the one-dimensional Bagnold model, the present numerical model achieves first-order convergence and gives better results than that based on the previous unified governing equations with the same numerical methods. In the simulation of the free drop of a water patch in a closed tank, a similar pressure curve with other reported work is given.

KEYWORDS

compressibility, governing equations, pressure-based method, THINC scheme, two-phase flow, WENO scheme

1 | INTRODUCTION

In some two-phase flow applications, for example, water–air interaction, one phase can often be considered as incompressible, while the other's compressibility cannot be neglected. Some previous studies[1-3] involving such flows used the Tait equation of state (EOS) to describe the behavior of nearly incompressible fluids. The Tait EOS leads to a stiff pressure law and a large sound speed in the liquid region, resulting in severe time step restriction. To circumvent the Tait EOS, we treat the nearly incompressible fluid as absolutely incompressible, that is, the density of that fluid is constant. The models for compressible flow and incompressible flow, however, are of disparate partial differential equations. The two models can be thought of as the basic problems for hyperbolic and elliptic partial differential equations.[4]

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To solve the conflict, a coupling strategy has been developed.[5-8] The idea is that the flows in the compressible and incompressible fluid are described by different simplified mathematical models, in which the governing equations are solved by different numerical methods. Usually, extrapolation of some characteristic variables across the interface is needed. This procedure may become complex or even impossible under certain violent interface evolution. Alternatively, some studies use one set of governing equations. Some of them employ the compressible flow solvers to tackle the two-phase flows. Others have developed strategies for two-phase flows from the unified methods for all speeds.[9,10] Most of these unified methods use the energy equation in terms of the time derivative of the pressure. Thus, the governing equations include an independent equation for pressure. In this article, we propose a novel pressure-based numerical scheme as a further simplification of the two-phase unified methods. This scheme is based on the following two assumptions.

(1) Incompressible liquid is the dominant fluid. Under some circumstances, for example, water and air, the density ratio is large; the liquid plays a leading role in the flow motion. For incompressible flows, pressure-based methods have been developed, such as marker-and-cell (MAC) method,[11] semi-implicit method for pressure-linked equations (SIMPLE) method,[12] and pressure-implicit with splitting of operators (PISO) algorithm.[13] The pressure-based method exploits the free divergence property of incompressible flow, solving the pressure field to maintain a divergence-free velocity field at each time step. When the incompressible or weakly compressible fluid dominates, rather than use a solver originally specified for compressible flows to tackle compressible–incompressible two-phase flows,[3,14] we choose to develop a pressure-based method.
(2) The density of gas only depends on pressure. For ideal gas, when the temperature variation is trifling, the density is a function of pressure only. Previous studies[15,16] about two-phase flows indicated that a small disturbance in density will cause a sharp pressure variation for weakly compressible liquids. The Tait EOS hampers an accurate numerical calculation of pressure. To overcome the difficulty, an independent pressure equation is deduced from the EOS and the mass conservation equation. Thus, there are two individual equations for density and pressure. The relationship between density and pressure, however, is obscured, because density and pressure are solved independently but are also linked through the EOS. Here, we explicitly take pressure as one primary variable, consider density as a function of pressure only, and exclude the density equation from the governing equations.

The high-order accurate weighted essentially nonoscillatory (WENO) schemes[17,18] are robust and have the advantage that the parameters do not required to be tuned, giving the convenience to verify the scheme proposed. In this article, we use the fifth-order WENO scheme to solve the advection phases of velocity and pressure. Alongside the WENO scheme for advection term, the third-order accurate total variation diminishing (TVD) Runge–Kutta (RK) method[17] is adopted for temporal integration.

The tangent of hyperbola for interface capturing (THINC) scheme, a VOF-type surface tracking method, was proposed by Xiao et al.[19] and later the multidimensional algebraic was improved.[20] As a geometrical-reconstruction-free method, this scheme is easy and simple to implement. The technique shows superiority in dealing with interface capturing, and has been validated in previous studies.[16,21,22] Here we employ the THINC scheme to solve the advection phase of gas volume fraction. The THINC scheme, like other VOF-type methods, ensures limited mass loss.

Section 2 concentrates on the formula derivation. Section 3 gives the details of numerical solutions, including the fractional step approach, WENO scheme, and THINC scheme. Section 4 is dedicated to validating the scheme proposed. The generalized Bagnold model is chosen as a benchmark test for one-dimensional two-phase flows, and the idealized free drop of a water volume in a closed tank as the two-dimensional test. An error comparison and convergence study of the present governing equations with the previous unified governing equation is also conducted. The last section gives the conclusions.

2 | GOVERNING EQUATIONS

The governing equations for two-phase flows are customarily written in terms of volume fractions. For the purpose of theoretical derivation, we consider only the one-dimensional inviscid flow without source terms. Here we consider one fluid as gas, the other as liquid. The individual continuity equations for the gas and liquid phases are

$$\frac{\partial \tilde{\rho}_g \alpha_g}{\partial t} + \frac{\partial \tilde{\rho}_g \alpha_g u}{\partial x} = 0, \tag{1}$$

$$\frac{\partial \tilde{\rho}_l \alpha_l}{\partial t} + \frac{\partial \tilde{\rho}_l \alpha_l u}{\partial x} = 0, \tag{2}$$

and the global momentum equation is

$$\frac{\partial \rho u}{\partial t} + \frac{\partial (\rho u^2 + p)}{\partial x} = 0. \tag{3}$$

In the equations above, the mixture density is defined as

$$\rho = \alpha_g \tilde{\rho}_g + \alpha_l \tilde{\rho}_l. \tag{4}$$

In the equations above, ρ is the density of the mixture, u is the velocity, p is the pressure. $\tilde{\rho}_g$ and $\tilde{\rho}_l$ are the individual phasic densities of gas and liquid, defined as mass of the phase per unit volume occupied by that phase. α_g and α_l are the volume fractions of the gas phase and the liquid phase. Note that $\alpha_g + \alpha_l = 1$.

The system is then closed by the phasic equations of state

$$\tilde{\rho}_g = \tilde{\rho}_g(p) \quad \text{and} \quad \tilde{\rho}_l = \tilde{\rho}_l(p). \tag{5}$$

The overall continuity equation is

$$\frac{\partial \rho}{\partial t} + \frac{\partial \rho u}{\partial x} = 0. \tag{6}$$

From Equations (3) and (6), one has

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = -\frac{1}{\rho} \frac{\partial p}{\partial x}. \tag{7}$$

With the liquid density, $\tilde{\rho}_l$, as constant, Equation (2) becomes

$$\frac{\partial \alpha_l}{\partial t} + \frac{\partial \alpha_l u}{\partial x} = 0. \tag{8}$$

Since $\alpha_g + \alpha_l = 1$, one has

$$\frac{\partial \alpha_g}{\partial t} + \frac{\partial \alpha_g u}{\partial x} = \frac{\partial u}{\partial x}, \tag{9}$$

which can be rewritten as

$$\frac{\partial \alpha_g}{\partial t} + u \frac{\partial \alpha_g}{\partial x} = (1 - \alpha_g) \frac{\partial u}{\partial x}. \tag{10}$$

Rewrite Equation (1) as

$$\alpha_g \frac{\partial \tilde{\rho}_g}{\partial t} + u \alpha_g \frac{\partial \tilde{\rho}_g}{\partial x} + \tilde{\rho}_g \left(\frac{\partial \alpha_g}{\partial t} + \frac{\partial \alpha_g u}{\partial x} \right) = 0. \tag{11}$$

Using Equation (9), Equation (11) reduces to

$$\alpha_g \frac{\partial \tilde{\rho}_g}{\partial t} + u \alpha_g \frac{\partial \tilde{\rho}_g}{\partial x} + \tilde{\rho}_g \frac{\partial u}{\partial x} = 0. \tag{12}$$

Assuming no temperature variation, the EOS for the gas can be written as $\tilde{\rho}_g = \tilde{\rho}_g(p)$. The sound speed of gas, $C_g = \sqrt{dp/d\tilde{\rho}_g}$, determines the relationship between pressure and density. Applying the EOS to Equation (12), the pressure

equation can be expressed as

$$\alpha_g \frac{\partial p}{\partial t} + u\alpha_g \frac{\partial p}{\partial x} + C_g^2 \tilde{\rho}_g \frac{\partial u}{\partial x} = 0. \tag{13}$$

Now we have a new set of equations

$$\frac{\partial \alpha_g}{\partial t} + u\frac{\partial \alpha_g}{\partial x} = (1-\alpha_g)\frac{\partial u}{\partial x}, \tag{14a}$$

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = -\frac{1}{\rho}\frac{\partial p}{\partial x}, \tag{14b}$$

$$\alpha_g \frac{\partial p}{\partial t} + u\alpha_g \frac{\partial p}{\partial x} = -C_g^2 \tilde{\rho}_g \frac{\partial u}{\partial x}. \tag{14c}$$

Noting that the left-hand side of Equation (14c) is not in conformance with the advection equation, we assume that α_g is a very small positive number where there is no gas.

Divided by α_g, Equation (14c) yields

$$\frac{\partial p}{\partial t} + u\frac{\partial p}{\partial x} = -C_g^2 \frac{\tilde{\rho}_g}{\alpha_g}\frac{\partial u}{\partial x}. \tag{15}$$

Later we can see that this strategy is practical, since the right-hand side of Equation (15) is never explicitly used to update pressure in a pressure projection method.

Theoretically, for the region occupied by incompressible fluid, there is

$$\alpha_g \frac{\partial p}{\partial t} + u\alpha_g \frac{\partial p}{\partial x} \equiv -C_g^2 \tilde{\rho}_g \frac{\partial u}{\partial x} \tag{16}$$

since $\alpha_g \equiv 0$ and $\partial u/\partial x \equiv 0$ where there is no gas. This means that Equation (14c) remains valuable for the compressible region, indicating that this strategy works for purely compressible or incompressible flows.

For one-dimensional inviscid compressible–incompressible two-phase flows, the final governing equations are

$$\frac{\partial \alpha_g}{\partial t} + u\frac{\partial \alpha_g}{\partial x} = (1-\alpha_g)\frac{\partial u}{\partial x}, \tag{17a}$$

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = -\frac{1}{\rho}\frac{\partial p}{\partial x}, \tag{17b}$$

$$\frac{\partial p}{\partial t} + u\frac{\partial p}{\partial x} = -C_g^2 \frac{\tilde{\rho}_g}{\alpha_g}\frac{\partial u}{\partial x}. \tag{17c}$$

Since $1-\alpha_g \equiv 0$ where there is no liquid, and $\partial u/\partial x \equiv 0$ where there is no gas, theoretically the right-hand side of Equation (17a) is constantly zero. Nevertheless we choose to keep it as a correction. In numerical simulation, α_g is neither 0 nor 1 but some number between 0 and 1 near the interface.

Assuming that the entropy is constant, the EOS for ideal gas is

$$p = p(\rho) \equiv C\rho^\gamma, \quad C \equiv \text{constant}, \tag{18}$$

where γ is the adiabatic constant depending on the gas considered.

Thus the sound speed C_g can be expressed as

$$C_g^2 = \frac{dp}{d\tilde{\rho}_g} = \gamma C \tilde{\rho}_g^{\gamma-1} = \gamma \frac{C\tilde{\rho}_g^\gamma}{\tilde{\rho}_g} = \gamma \frac{p}{\tilde{\rho}_g}. \tag{19}$$

Using Equation (19), Equation (17c) further reduces to

$$\frac{\partial p}{\partial t} + u\frac{\partial p}{\partial x} = -\frac{\gamma p}{\alpha_g}\frac{\partial u}{\partial x}, \tag{20}$$

which is more practical.

Similarly, the governing equations for multidimensional compressible–incompressible two-phase flows are

$$\frac{\partial \alpha_g}{\partial t} + u_i \frac{\partial \alpha_g}{\partial x_i} = (1 - \alpha_g) \frac{\partial u_i}{\partial x_i}, \tag{21a}$$

$$\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j} = \frac{1}{\rho} \frac{\partial \tau_{ij}}{\partial x_j} - \frac{1}{\rho} \frac{\partial p}{\partial x_i} + f_i, \tag{21b}$$

$$\frac{\partial p}{\partial t} + u_i \frac{\partial p}{\partial x_i} = -C_g^2 \frac{\tilde{\rho}_g}{\alpha_g} \frac{\partial u_i}{\partial x_i}, \tag{21c}$$

where u_i is the velocity, f_i is the external force, and $\tau_{ij} = 2\mu S_{ij} - 2\mu \delta_{ij} S_{kk}/3$ is the viscous part of the stress tensor for Newtonian fluid, where $S_{ij} = (\partial u_i/\partial x_j + \partial u_j/\partial x_i)/2$ is the rate of strain tensor and δ_{ij} denotes the Kronecker's symbol. The viscosity μ should be defined as $\mu = \alpha_g \mu_g + \alpha_l \mu_l$, in which μ_g and μ_l are the viscosity of the gas and the liquid involved. Since Equation (21c) is based on the inviscid gas assumption, μ_g here is set to zero and one has $\mu = \alpha_l \mu_l$.

Equation (21c) can also be rewritten as

$$\frac{\partial p}{\partial t} + u_i \frac{\partial p}{\partial x_i} = -\frac{\gamma p}{\alpha_g} \frac{\partial u_i}{\partial x_i}. \tag{22}$$

The following are the governing equations for water–air two-phase flow based on Yabe et al.'s unified procedure method.[15,16]

$$\frac{\partial \alpha}{\partial t} + u_i \frac{\partial \alpha}{\partial x_i} = 0, \tag{23a}$$

$$\frac{\partial \rho}{\partial t} + u_i \frac{\partial \rho}{\partial x_i} = -\rho \frac{\partial u_i}{\partial x_i}, \tag{23b}$$

$$\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j} = \frac{1}{\rho} \frac{\partial \tau_{ij}}{\partial x_j} - \frac{1}{\rho} \frac{\partial p}{\partial x_i} + f_i, \tag{23c}$$

$$\frac{\partial p}{\partial t} + u_i \frac{\partial p}{\partial x_i} = -\rho C_S^2 \frac{\partial u_i}{\partial x_i}, \tag{23d}$$

where C_S is the sound speed in the mixture, α is the volume fraction of gas or liquid, and the rest variables are of the same meaning in Equation (21).

We here call Equation (21) the present governing equations (GQ) and Equation (23) the previous governing equations (GQ). The main differences between the two sets of governing equations are that 1) Equation (21) does not contain the equation for density; 2) the right-hand side of Equation (21c) utilize the variables only related to gas, while that of Equation (23d) takes variables of the mixture. The present GQ, thus, can be seen as a simplification of the previous GQ. In the previous GQ, the density and pressure are respectively calculated via a separate equation, that is, Equations (23b) and (23d), but the two are also related through the EOS. As pointed in Fedkiw et al.'s Ghost Fluid Method,[1] the pressure is continuous across the interface, which allows to solve the pressure equation with high accuracy. We here choose to omit the density equation. The constant liquid density and the continuous pressure supplemented by the EOS for gas naturally preserve the density discontinuity across the interface.

The cut of computational consumption from the exclusion of the density equation is trivial compared with that to solve the Poisson equation to obtain the pressure field. That the density is derived from the pressure and the EOS instead of via the advection Equation (23b), however, results in a more accurate density profile as shown later in the article.

3 | NUMERICAL SOLUTIONS

3.1 | The fractional step approach

The numerical solution of Equation (21) can be divided into three steps using the fractional step approach.[23,24]

(1) Advection phase

$$\frac{\alpha_g^* - \alpha_g^n}{\Delta t} + u_i^n \frac{\partial \alpha_g^n}{\partial x_i} = 0, \tag{24}$$

$$\frac{u_i^* - u_i^n}{\Delta t} + u_j^n \frac{\partial u_i^n}{\partial x_j} = 0, \tag{25}$$

$$\frac{p^* - p^n}{\Delta t} + u_i^n \frac{\partial p^n}{\partial x_i} = 0. \tag{26}$$

Equation (24) is computed by the THINC scheme, which is specified for the advection equation of the volume of fraction function. Equations (25) and (26) are calculated by the WENO scheme. α_g^*, u^*, and p^* are the intermediate values obtained.

(2) Nonadvection phase I

The second intermediate values, u_i^{**}, are obtained by

$$\frac{u_i^{**} - u_i^*}{\Delta t} = \frac{2\mu}{\rho} \frac{\partial}{\partial x_j} \left(S_{ij}^* - \frac{1}{3} \delta_{ij} S_{kk}^* \right) + f_i. \tag{27}$$

(3) Nonadvection phase II

$$\frac{\alpha_g^{n+1} - \alpha_g^*}{\Delta t} = (1 - \alpha_g^*) \frac{\partial u_i^{n+1}}{\partial x_i}, \tag{28}$$

$$\frac{u_i^{n+1} - u_i^{**}}{\Delta t} = -\frac{1}{\rho^*} \frac{\partial p^{n+1}}{\partial x_i}, \tag{29}$$

$$\frac{p^{n+1} - p^*}{\Delta t} = -C_g^2 \frac{\tilde{\rho}_g^*}{\alpha_g^*} \frac{\partial u_i^{n+1}}{\partial x_i}. \tag{30}$$

Taking divergence of Equation (29) and substituting $\partial u_i^{n+1}/\partial x_i$ by Equation (30), the pressure equation is obtained,

$$\frac{\alpha_g^*(p^{n+1} - p^*)}{C_g^2 \tilde{\rho}_g^* \Delta t^2} + \frac{1}{\Delta t} \frac{\partial u_i^{**}}{\partial x_i} = \frac{\partial}{\partial x_i} \left(\frac{1}{\rho^*} \frac{\partial p^{n+1}}{\partial x_i} \right), \tag{31}$$

which is a Poisson-type equation. Central differencing is used to discretize Equation (31), which leads to a coupled linear system of equations for the unknown pressure values p^{n+1}. This system is solved via the direct LDLt factorization using the high-level C++ library Eigen.

Now the gas volume fraction, α_g, is in the numerator rather than in the denominator. As a result, that $\alpha_g = 0$ makes sense. In the region occupied by incompressible fluid, $\alpha_g \equiv 0$, Equation (31) reduces to

$$\frac{1}{\Delta t} \frac{\partial u_i^{**}}{\partial x_i} = \frac{\partial}{\partial x_i} \left(\frac{1}{\rho^*} \frac{\partial p^{n+1}}{\partial x_i} \right), \tag{32}$$

which is the conventional Poisson equation for incompressible flows.

After Equation (31) is solved over the computation domain, the pressure p^{n+1} is obtained. Then u_i^{n+1} and α_g^{n+1} can be computed by Equations (29) and (28).

For a staggered grid, density at the cell boundary is estimated by the following weighted average.[25]

$$\rho_{i+1/2} \approx \frac{\rho_i \rho_{i+1} (\rho_i + \rho_{i+1})}{\rho_i^2 + \rho_{i+1}^2}. \tag{33}$$

The above estimation is used in Equations (31) and (29).

The CFL time step restriction is

$$\alpha = \Delta t \left(\frac{\max\{|\boldsymbol{u}|\}}{\min\{\Delta x, \Delta y\}} \right), \tag{34}$$

where α is the CFL number. Here the time-step restriction on the viscous term is omitted since α is practically rather small to achieve a high accuracy in the gas phase.

3.2 | WENO scheme

The advection phase equations in Equations (21b), (21c) share the same form

$$\frac{\partial \phi}{\partial t} + u_i \frac{\partial \phi}{\partial x_i} = 0. \tag{35}$$

The one-dimensional version of Equation (35) is

$$\frac{\partial \phi}{\partial t} + u \frac{\partial \phi}{\partial x} = 0. \tag{36}$$

The WENO scheme is used to evaluate the $\partial\phi/\partial x$ term numerically. On grid point x_i, the sign of u_i determines whether the upwind difference ϕ_x^+ or the backwind difference ϕ_x^- is used to approximate $\partial\phi/\partial x$. If $u_i > 0$, $\partial\phi/\partial x$ is approximated with ϕ_x^-. If $u_i < 0$, $\partial\phi/\partial x$ is approximated with ϕ_x^+. If $u_i = 0$, the advection term vanished. Here gives a brief introduction to the Hamilton-Jabobi WENO (HJ WENO) scheme.

A subset of $\{\phi_{i-3}, \phi_{i-2}, \phi_{i-1}, \phi_i, \phi_{i+1}, \phi_{i+2}\}$ is used to calculate $(\phi_x^-)_i$. There are three possible third-order accurate approximations to $(\phi_x^-)_i$. Define $v_1 = (\phi_{i-2} - \phi_{i-3})/\Delta x$, $v_2 = (\phi_{i-1} - \phi_{i-2})/\Delta x$, $v_3 = (\phi_i - \phi_{i-1})/\Delta x$, $v_4 = (\phi_{i+1} - \phi_i)/\Delta x$, $v_5 = (\phi_{i+2} - \phi_{i+1})/\Delta x$. The three possible approximations are given as

$$\phi_x^1 = \frac{v_1}{3} - \frac{7v_2}{6} + \frac{11v_3}{6}, \tag{37a}$$

$$\phi_x^2 = -\frac{v_2}{3} + \frac{5v_3}{6} + \frac{v_4}{3}, \tag{37b}$$

$$\phi_x^3 = \frac{v_3}{3} + \frac{5v_4}{6} - \frac{v_5}{6}. \tag{37c}$$

The HJ WENO approximation of $(\phi_x^-)_i$ is a convex combination of the three approximations given by

$$\phi_x^- = \omega_1 \phi_x^1 + \omega_2 \phi_x^2 + \omega_3 \phi_x^3, \tag{38}$$

where the $0 \le \omega_k \le 1$ are the weights and there is $\omega_1 + \omega_2 + \omega_3 = 1$.

According the work of Jiang and Peng,[26] the weights are defined as following.

Let

$$S_1 = \frac{13}{12}(v_1 - 2v_2 + v_3)^2 + \frac{1}{4}(v_1 - 4v_2 + 3v_3)^2, \tag{39a}$$

$$S_2 = \frac{13}{12}(v_2 - 2v_3 + v_4)^2 + \frac{1}{4}(v_2 - v_4)^2, \tag{39b}$$

$$S_3 = \frac{13}{12}(v_3 - 2v_4 + v_5)^2 + \frac{1}{4}(3v_3 - 4v_4 + v_5)^2, \tag{39c}$$

and

$$\alpha_1 = \frac{0.1}{(S_1 + \epsilon)^2}, \quad \alpha_2 = \frac{0.6}{(S_2 + \epsilon)^2}, \quad \alpha_3 = \frac{0.3}{(S_3 + \epsilon)^2} \tag{40}$$

with $\epsilon = 10^{-6} \max\{v_1^2, v_2^2, v_3^2, v_4^2, v_5^2\} + 10^{-99}$ to avoid division by zero.

Normalize the α_k to obtain ω_k

$$\omega_k = \frac{\alpha_k}{\alpha_1 + \alpha_2 + \alpha_3}, \quad k = 1, 2, 3. \tag{41}$$

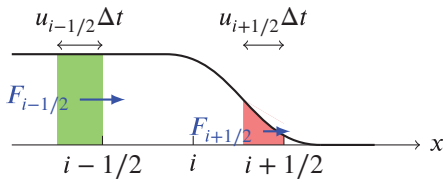

FIGURE 1 Schematic figure for the one-dimensional THINC scheme [Colour figure can be viewed at wileyonlinelibrary.com]

$(\phi_x^+)_i$ is constructed with a subset of $\{\phi_{i-2}, \phi_{i-1}, \phi_i, \phi_{i+1}, \phi_{i+2}, \phi_{i+3}\}$. Define $v_1 = (\phi_{i+3} - \phi_{i+2})/\Delta x$, $v_2 = (\phi_{i+2} - \phi_{i+1})/\Delta x$, $v_3 = (\phi_{i+1} - \phi_i)/\Delta x$, $v_4 = (\phi_i - \phi_{i-1})/\Delta x$, $v_5 = (\phi_{i-1} - \phi_{i-2})/\Delta x$ and the remaining procedures are the same with those to calculate $(\phi_x^-)_i$. More details are shown in Sun's work.[18]

3.3 | THINC scheme

Here the THINC scheme is employed to capture the interface. Though the original one-dimensional THICN scheme[19] is devised for the advection equation of the VOF function

$$\frac{\partial \phi}{\partial t} + \frac{\partial}{\partial x}(u\phi) - \phi\frac{\partial u}{\partial x} = 0, \tag{42}$$

which is not of the same form with the advection phase of Equation (17a), the two are actually identical.

For mesh cell $[x_{i-1/2}, x_{i+1/2}]$ at $t = t^n$, the cell-averaged values of ϕ is given as

$$\overline{\phi}_i^n = \frac{1}{\Delta x_i}\int_{x_{i-1/2}}^{x_{i+1/2}} \phi(x, t^n)dx \tag{43}$$

with $\Delta x = x_{i+1/2} - x_{i-1/2}$. The solution at $t = t^{n+1}$ is obtained by the finite volume formulation

$$\overline{\phi}_i^{n+1} = \overline{\phi}_i^n - \left(F_{i+1/2} - F_{i-1/2}\right)/\Delta x + \overline{\phi}_i^n\left(u_{i+1/2} - u_{i-1/2}\right)\Delta t/\Delta x, \tag{44}$$

where $F_{i+1/2}$ is the flux transported across boundary $x = x_{i+1/2}$ during $\Delta t = t^{n+1} - t^n$ demonstrated in Figure 1 and is computed by

$$F_{i+1/2} = \begin{cases} -\int_{t^n}^{t^{n+1}} \boldsymbol{\Phi}_i\left(x_{i+1/2} - u_{i+1/2}(t - t^n)\right)dt, & \text{if } u_{i+1/2} \geq 0 \\ \int_{t^n}^{t^{n+1}} \boldsymbol{\Phi}_{i+1}\left(x_{i+1/2} - u_{i+1/2}(t - t^n)\right)dt, & \text{if } u_{i+1/2} < 0 \end{cases}, \tag{45}$$

where $\boldsymbol{\Phi}_i(x)$ is the reconstructed interpolation function.

The THINC scheme uses the piecewise hyperbolic tangent function

$$\boldsymbol{\Phi}_i(x) = \frac{1}{2}\left(1 + \gamma\tanh\left(\beta\left(\frac{x - x_{i-1/2}}{\Delta x_i} - \tilde{x}_i\right)\right)\right), \tag{46}$$

where $\gamma = 1$ for $\phi_{i-1} < \phi_{i+1}$ and $\gamma = -1$ for $\phi_{i-1} > \phi_{i+1}$. β is the parameter to control the slope and thickness of the interface jump. Following Xiao's work,[20] $\beta = 2.3$ is set for all simulation cases in this article. With $\beta = 2.3$, the jump can be resolved by approximately two to three mesh cells. $\tilde{x}_i$ is the jump center and is given by solving

$$\frac{1}{\Delta x}\int_{x_{i-1/2}}^{x_{i+1/2}} \boldsymbol{\Phi}_x(x) = \overline{\phi}_i^n. \tag{47}$$

The explicit formula to calculate the flux can be found in Xiao's work.[20] Though it is suggested by Xiao that the operator splitting can be used to conduct the multidimensional computation, an improved multidimensional THINC scheme[27] is used for the two-dimensional tests in present work.

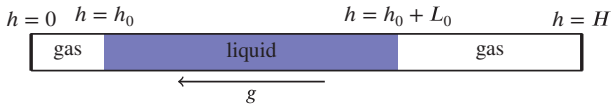

FIGURE 2 The initial conditions of the generalized Bagnold model, a closed tube at both ends, in which an incompressible liquid layer is trapped between two gas layers [Colour figure can be viewed at wileyonlinelibrary.com]

3.4 | Temporal integration

In this article, the third-order accurate TVD RK scheme is adopted to obtain a higher accuracy temporally. As pointed out in Reference 4, TVD RK schemes guarantee that so long as the forward Euler building block produces no spurious oscillations, this higher-order accurate temporal integration also produces no spurious oscillations.

The third-order accurate TVD RK scheme proposed in Reference 17 reads as follows:

$$\phi^{(1)} = \phi^n + \Delta t \mathcal{L}(\phi^n), \tag{48a}$$

$$\phi^{(2)} = \frac{3}{4}\phi^n + \frac{1}{4}\phi^{(1)} + \mathcal{L}(\phi^{(1)}), \tag{48b}$$

$$\phi^{n+1} = \frac{1}{3}\phi^n + \frac{2}{3}\phi^{(2)} + \frac{2}{3}\mathcal{L}(\phi^{(2)}), \tag{48c}$$

where ϕ is the primary variable (α_g, u_i, or p), $\mathcal{L}$ is the respective operator evaluating the numerically approximated RHS of Equation (21). The superscripts (1) and (2) are substeps between the present and subsequent time steps denoted by n and $n+1$.

4 | NUMERICAL TESTS

4.1 | Generalized Bagnold model

Bagnold[28] studied the problem of a solid piston (incompressible liquid) sliding along a tube. In the original work of Bagnold, the tube was open at one end and closed at the other. The liquid piston was moving toward the wall at an initial velocity with ideal gas trapped between the wall and the piston. Here we consider a generalized Bagnold model,[29] demonstrated in Figure 2. The liquid piston has a uniform initial velocity for $x \in [h_0, h_0 + L_0]$. The initial condition is a compatible linear distribution of velocity in the gas layer. Meanwhile, a uniform and constant external force, for example, gravity, is exerted on the liquid and the gas.

4.1.1 | Analytical solution

The dimensionless formulation and an analytical solution of the generalized Bagnold model are given by Brosset et al.[29] Here presents a brief introduction. The dimensionless height of the left gas pocket, $\tilde{h} = h/h_0$, follows

$$\tilde{h}''(\tilde{t}) = -\tilde{S}_g + \tilde{h}(\tilde{t})^{-\tilde{\gamma}} - \left(1 + \frac{1}{\tilde{C}}(1 - \tilde{h}(\tilde{t}))\right)^{-\tilde{\gamma}} \tag{49}$$

with the initial conditions: $\tilde{h}(0) = 1$; $\tilde{h}'(0) = \mathrm{sgn}(u_0)\sqrt{\tilde{S}_B}$.

When the liquid is incompressible, the problem is entirely characterized by four dimensionless numbers:

- aspect ratio $\tilde{C}$, given by $\tilde{C} = (H - L_0 - h_0)/h_0$
- impact number related to body force $\tilde{S}_g$, given by $\tilde{S}_g = (\rho_l L_0 g)/p_0$
- impact number related to the initial velocity $\tilde{S}_B$, given by $\tilde{S}_B = (\rho_l L_0 u_0^2)/(p_0 h_0)$
- compressibility of the gas $\tilde{\gamma}$, given by $\tilde{\gamma} = \gamma$

TABLE 1 Description of the cases

Case	Scale	Liquid	Gas	Pressure (Pa)	S_g	S_B
1	1:1	Water	Air	100,000	0.784	0.0
2	1:1	LNG	NG	100,000	0.357	0.0
3	1:40	LNG	NG	100,000	0.00892	0.0
4	1:40	1:40-scaled LNG	1:40-scaled NG	2500	0.357	0.0
5	1:1	Water	Air	100,000	0.0	2.0

TABLE 2 Properties of the fluids

Fluid (liquid)	γ	ρ_0 (kg/m^3)	Fluid (gas)	γ	ρ_0 (kg/m^3)
Water	/	1000	Air	1.4	1.0
LNG	/	455	NG	1.3	1.82
1:40-scaled LNG	/	455	1:40-scaled NG	1.3	1.82

where g is the uniform body force, ρ_l is the density of the liquid, u_0 is the initial velocity of the liquid, p_0 is the initial pressure, γ is the adiabatic constant of the gas, $\tilde{t} = t/\sqrt{(\rho_l L_0 h)/p_0}$ is the dimensionless time, and H, L_0, h_0 are lengths illustrated in Figure 2.

4.1.2 | Numerical results

The generalized Bagnold model has been widely chosen as a benchmark test.[30] In present work, several cases are studied with different properties of the fluids or different geometrical scales and lengths, either induced by body force or forced motion with an initial velocity. The calculation condition of the cases is given in Table 1, the properties of the fluids in Table 2. For Cases 1–4, the lengths are $h_0 = 2$ m, $L_0 = 8$ m, $H = 15$ m; for Case 5, the lengths are $h_0 = 0.4$ m, $L_0 = 0.2$ m, $H = 1.0$ m. Case 1 has been studied in References 3,16; Cases 2–4 are under the same conditions in the work of Boeck et al.[31]

When subjected to a uniform body force $\mathbf{g}$, the piston would first accelerate and then decelerate. The piston would first move to the left, leaving the gas ahead of it compressed and the gas behind it expanded. The left gas pocket is diminishing and the pressure at $h = 0$ is increasing. On the other hand, the right pocket is expanding and the pressure at $h = H$ is decreasing. As a result, the pressure difference at the two ends of the liquid piston grows bigger enough to reverse the acceleration direction. When the piston travels the farthest to the left, the pressure at $h = 0$ reaches its peak, the pressure at $h = H$ its nadir. Then the piston moves back, starting a periodic motion.

Cases 1 and 2 are simulated with a constant time interval $\Delta t = 1$ ms, Cases 3 and 4 with a time interval $\Delta t = 1/40$ ms to keep the same CFL condition. All cases are computed with 300 uniform meshes. We rescale the data according to the Froude scaling and plot all the results in Figure 3. As expected, Cases 2 and 4 yield identical results. A multicycle running of Case 1 is performed. The pressures and densities of both ends are recorded. Figures 3 and 4 show that pressures and densities change periodically. The numerical results are in good accordance with analytical solutions, indicating that the scheme works well for one-dimensional compressible–incompressible flows driven by body force.

The initial velocity of the liquid is 20 m/s to the right in Case 5. Like body-force-induced piston, piston with an initial velocity also oscillate periodically. Case 5 is simulated to $t = 0.1$ s with 200, 400, 800 uniform grid cells. The CFL number is set equal to 0.01. The results are illustrated in Figure 5. As pointed in Reference 29, the velocity is constant in the liquid and distributes linearly in the gas, whereas the pressure is constant in the gas but distributes linearly in the liquid. Figure 5 shows that this characteristic as well as the density jump at the interface are well preserved. The simulation results are nearly identical with analytical solution with 800 grid cells as shown in Figure 5(C). It means that the derived governing equations can accurately reflect the physics in the generalized Bagbold model.

Case 5 is also simulated by the previous governing equations with the same numerical methods described in Section 3. The error of pressure is chosen to carry out the convergence study with the number of cells $N = 100$, 200, 400, 800, and

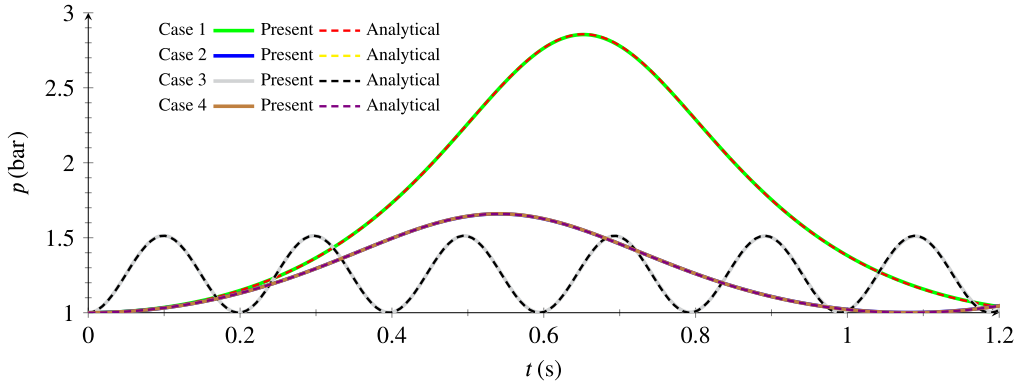

FIGURE 3 Pressure histories at $h = 0$ of Cases 1–4 [Colour figure can be viewed at wileyonlinelibrary.com]

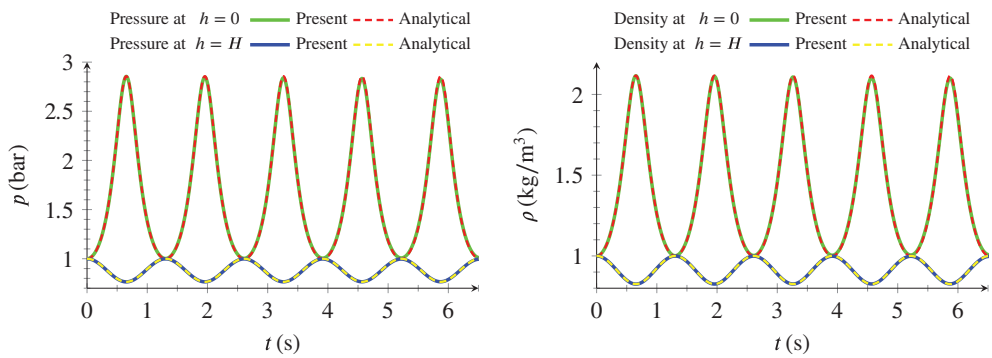

FIGURE 4 Multicycle results of Case 1, comparison of gas pressure at $h = 0$ and $h = H$ (left), comparison of gas density at $h = 0$ and $h = H$ (right) [Colour figure can be viewed at wileyonlinelibrary.com]

1600. The L_1, L_2, and L_∞ errors of pressure at $t = 0.1$ s are measured by

$$E(L_1) = \frac{1}{N}\sum_{i=1}^{N}\frac{|p^{\text{num}} - p^{\text{ref}}|}{p_0}, \tag{50}$$

$$E(L_2) = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\frac{|p^{\text{num}} - p^{\text{ref}}|^2}{p_0^2}}, \tag{51}$$

$$E(L_\infty) = \max_i\left(\frac{|p^{\text{num}} - p^{\text{ref}}|}{p_0}\right), \tag{52}$$

where p denotes pressure; N is the cell number; p^{num} and p^{ref} are the simulation result and the analytical solution.

The convergence order is computed by

$$O = \frac{1}{\ln(2)}\ln\left(\frac{E^q}{E^{q+1}}\right), \tag{53}$$

where E^q is the error of the qth resolution.

Tables 3 and 4 give the errors and convergence order in the simulation of Case 5 with the two sets of governing equations. It shows that the present numerical scheme achieves first-order convergence and gives less error than that with previous governing equations in this one-dimensional case.

As shown in Figures 6 and 5(B), the variable profiles at $t = 0.1$ s give a more clear comparison of errors with the two set governing equations. Both cases are calculated with 400 uniform grid cells. The results in Figure 5(B) given by the

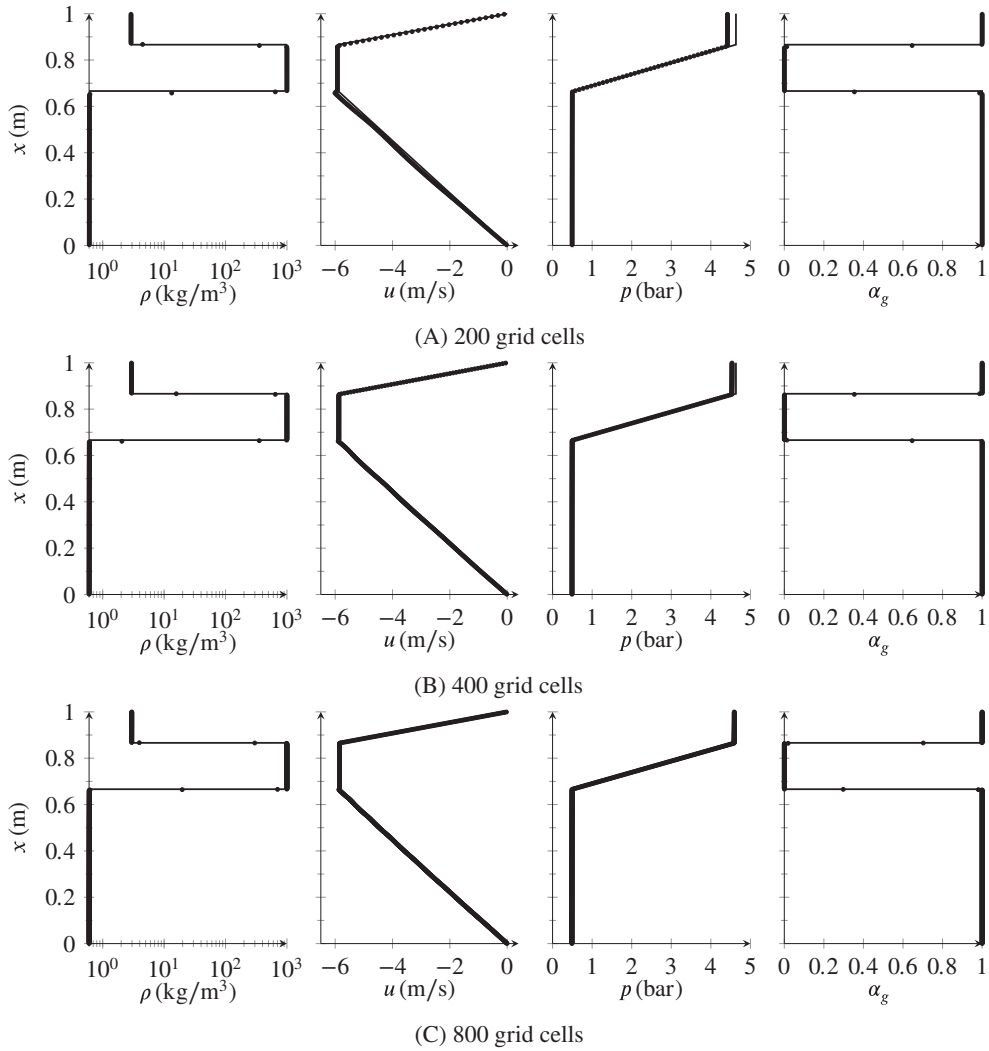

FIGURE 5 Variable distributions at $t = 0.1$ s of Case 5 with present governing equations. Lines are analytical solutions, and scatters are simulation results

TABLE 3 Errors and convergence orders with present governing equations

Number of cells	$E(L_1)$	$O(L_1)$	$E(L_2)$	$O(L_2)$	$E(L_\infty)$	$O(L_\infty)$
100	8.43e−02	–	1.68e−01	–	4.37e−01	–
200	4.08e−02	1.05	8.18e−02	1.03	2.17e−01	1.01
400	1.92e−02	1.09	3.86e−02	1.09	1.04e−01	1.06
800	8.21e−03	1.23	1.65e−02	1.23	4.62e−02	1.17
1600	3.22e−03	1.35	5.69e−03	1.53	1.74e−02	1.40

TABLE 4 Errors and convergence orders with previous governing equations

Number of cells	$E(L_1)$	$O(L_1)$	$E(L_2)$	$O(L_2)$	$E(L_\infty)$	$O(L_\infty)$
100	1.01e−01	–	1.73e−01	–	4.55e−01	–
200	1.59e−02	2.66	2.59e−02	2.74	1.29e−01	1.82
400	1.24e−02	0.36	2.13e−02	0.29	8.80e−02	0.55
800	9.68e−03	0.35	1.79e−02	0.25	4.97e−02	0.82
1600	7.26e−03	0.42	1.38e−02	0.37	3.77e−02	0.40

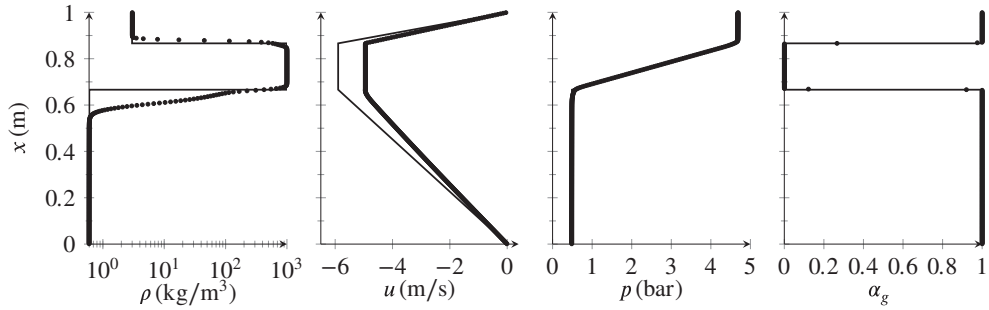

FIGURE 6 Variable distributions at $t = 0.1$ s of Case 5 with previous governing equations. Lines are analytical solutions, and scatters are simulation results

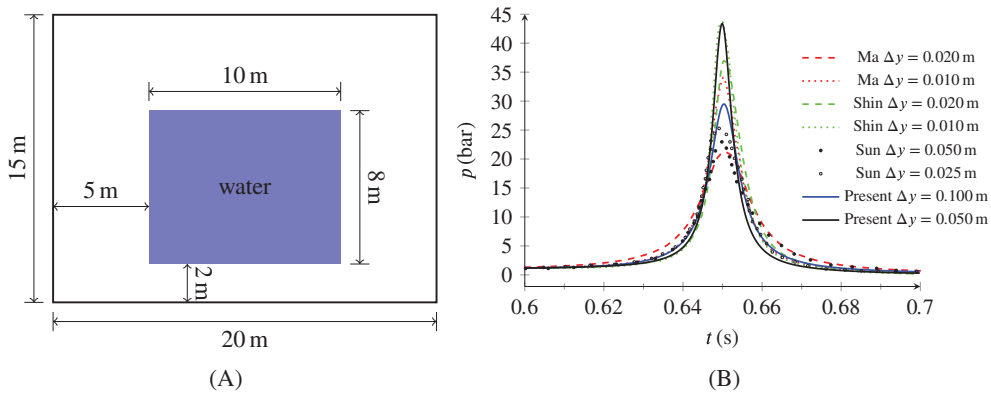

FIGURE 7 (A) Initial condition for the free drop of a water patch in a closed tank; (B) Comparison of pressure at the bottom center [Colour figure can be viewed at wileyonlinelibrary.com]

present GQ agree well with analytical solutions. The sharpness of density profile and gas volume profile is preserved as the interface is limited within two or three mesh cells. The jump thickness of density given by the previous GQ, as demonstrated in Figure 6, extends beyond a dozen mesh cells, which in turn leads to a larger error in velocity with the previous GQ. The discontinuity of density across the interface hinders a high accuracy in the numerical calculation of the advection term. Exclusion of the density equation and derivation of density from pressure and the EOS instead, thus, improve the simulation results.

With an Intel(R) Xeon(R) CPU E5-1620 v4 @ 3.50 GHz single core, the running times of Case 5 with 400 uniform meshes to $t = 0.1$ s with the previous GQ and the present GQ are 38.1386 and 38.5311 s. The cut of the density equation does not considerably bring less computing time, but with the same cost, the present GQ give much less error.

4.2 | Free drop of water patch

The idealized free drop of water patch in a closed tank is a benchmark test proposed in Reference 30. As illustrated in Figure 7(A), a 10 m-by-8 m water patch ($\rho = 1000$ kg/m^3) is initially at rest in the air ($\rho = 1$ kg/m^3, $\gamma = 1.4$). The water patch is 2-m high above the bottom and 5 m away from the left and right wall. The background pressure is 1 bar. Due to gravity, the water column will drop and hit the tank bottom. The pressure at the bottom center is an index of the impact violence.

Two sets of structured uniform grids are selected. The first one has 150 mesh cells equally distributed in the vertical direction and 200 in the horizontal direction; the second one has double cells along each direction. A constant CFL = 0.05 is set. Figures 8 and 9 give snapshots of volume fraction distributions, pressure distributions, and velocity fields in the tank at four different times. As pointed in Reference 3, a small portion of air is trapped between the water patch and the tank bottom during the drop. The air trapped is compressed to the minimum volume at about 0.65 s. As the pressure declines, the air expands and forms an air pocket. As the water patch evolves, the pocket changes too. There is an obvious

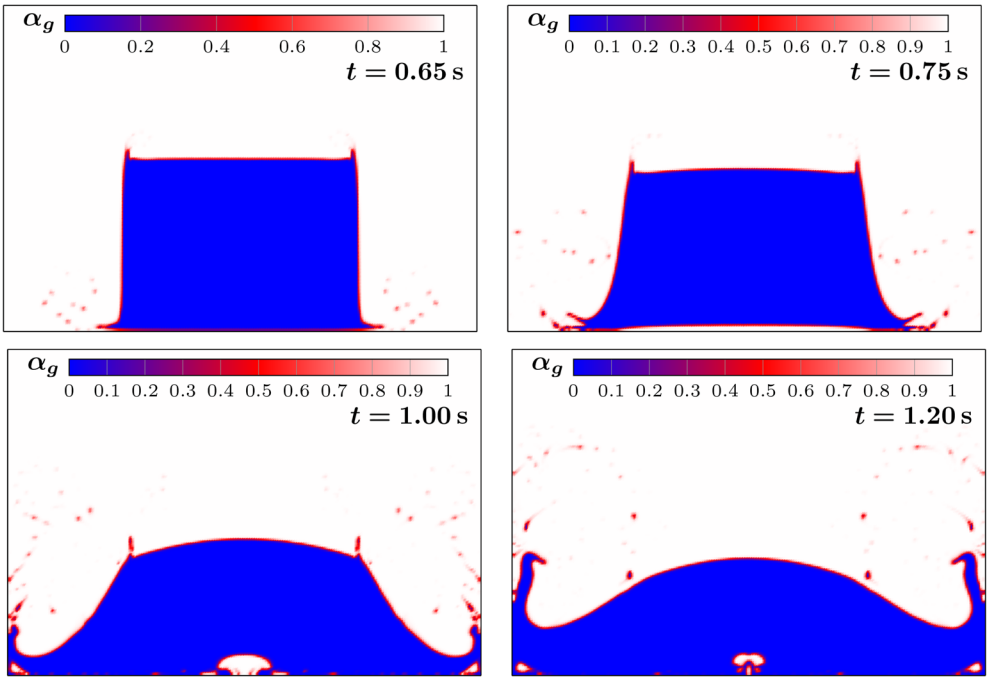

FIGURE 8 Distributions of the volume fraction at different times [Colour figure can be viewed at wileyonlinelibrary.com]

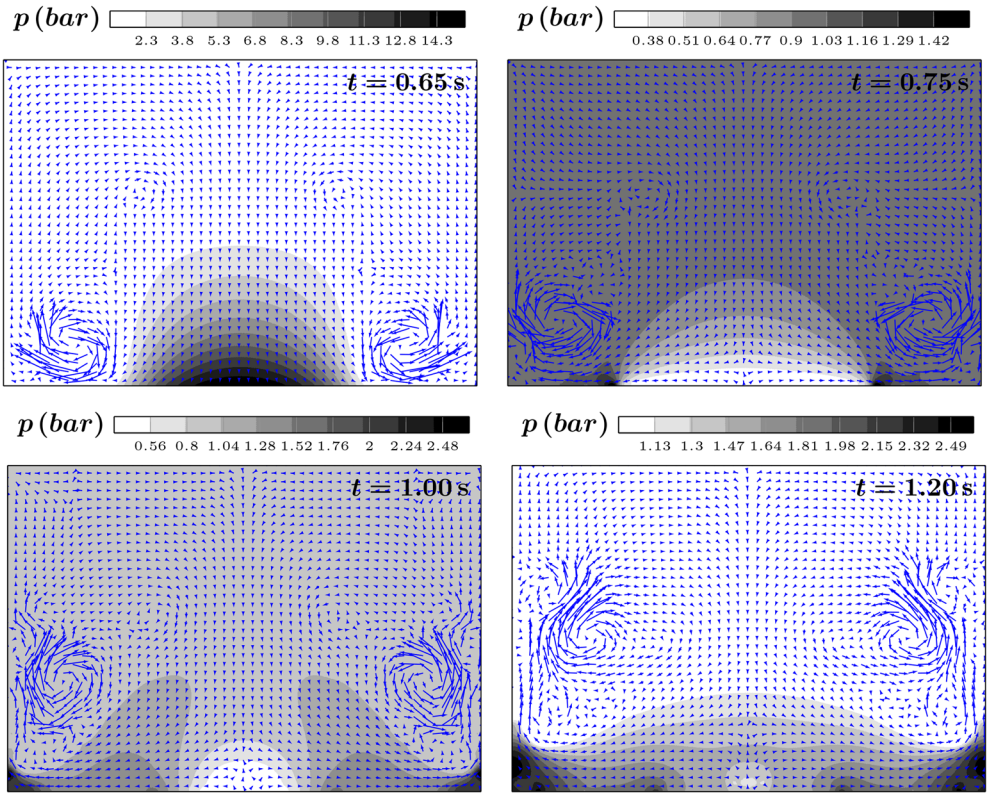

FIGURE 9 Pressure distributions and velocity fields at different times [Colour figure can be viewed at wileyonlinelibrary.com]

contraction of the air pocket in the later period as reflected in the bottom two figures in Figure 8. It is also clear that the pressure near the bottom center at $t = 1.20$ s is larger than that at $t = 1.00$ s, which explains the contraction. The same phenomenon has been observed in Shin's work.[25] The compression and expansion of the air pocket verify the scheme's capacity to take the gas's compressibility into consideration.

Figure 7(B) gives the comparison of the time history of pressure at the bottom center with the works of Ma et al.,[3] Sun et al.,[16] and Shin.[25] The peak pressure occurs at about 0.65 s. Among those works, the present peak value is closest to that of Shin's. Shin[25] uses a similar unified method, in which the pressure equation is omitted rather than the density equation. In addition, the WENO scheme is also adopted to solve the advection terms. The pressure falls after the peak, reaching a value of about 0.5 bar. The pressure after the peak agrees well with the result of other works. Despite the discrepancy in the peak value, the present work gives a good agreement in time and duration of the impact.

5 | CONCLUSIONS

In this article, a pressure-based numerical scheme for two-phase flows at low Mach numbers is presented. One phase is assumed to be incompressible liquid, whereas the other is considered to be compressible gas. Inspired by the pressure projection method used in incompressible flows, we develop a liquid-oriented method, in which the liquid is the dominant phase. The derived governing equations are simpler than the previous unified governing equations.

Numerical tests show that the results of present work are in good accordance with the analytical solutions and other's numerical results, which validates the scheme's capability in tackling incompressible–compressible two-phase flows. The general Bagnold model is simulated with the present and the previous governing equations. It shows that in this case the numerical model with present governing equations achieves first-order convergence, gives less diffusion in density and thus leads to better results. The simulation of the free drop of a water patch in a closed tank yields a similar time history of pressure at the bottom center with other numerical results. The compressibility of air is fully reflected.

Given that under certain circumstances the liquid's compressibility is neglectable but not the gas's, for example, air-water flows in coastal engineering, ocean engineering, and other fields, this scheme can further be applied to such realms.

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CONFLICT OF INTEREST
The authors declare no potential conflict of interests.

AUTHOR CONTRIBUTIONS
Zhilong Wei: Writing—original draft preparation, conceptualization, methodology, software, validation. Qin Jiang: Writing—review and editing, supervision, project administration, funding acquisition. Sihang Nie: Software, resources, formal analysis.

DATA AVAILABILITY STATEMENT
The data that support the findings of this study are available from the corresponding author upon reasonable request.

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